



🏆 LABLAB.AI ALPACA AI TRADING AGENTS HACKATHON • OPTIONS ALPHA AGENTS TRACK



AlgoPaca

Autonomous Quantitative Trading Desk on Alpaca's Complete Developer Stack

An institutional-grade, MIT-licensed trading platform built natively on **Alpaca Trading API**, **Model Context Protocol (MCP)**, and **Alpaca CLI**.
Features 6 quantitative engines, multi-provider LLM reasoning (Gemini, Claude, GPT, Grok, Featherless AI), options alpha overlays, precious metals macro intelligence, and crash-safe risk guardrails.

🔵 ⚡ Alpaca Trading API

🟣 🛠️ Built-In MCP Server

🟢 📄 Headless Alpaca CLI

🟡 📊 Options Alpha Overlay

🟢 🧠 Multi-LLM + Featherless AI

🔴 🛡️ 7 Pre-Trade Risk Gates

🟡 💰 \$100,000 Paper Balance

🟣 ⚖️ MIT Open Source

\$0 / mo

100% FREE & OPEN SOURCE

6 Engines

OPTIONS & EQUITY PRESETS

7 Gates

PRE-TRADE RISK CIRCUIT

API+MCP+CLI

TRIPLE ALPACA MODALITY

\$100,000

COMPETITION PAPER ACCOUNT



Retail Bots are Broken. AlgoPaca Restores Control.

Most retail trading bots offer black-box signals, opaque subscriptions, and fragile architectures that drop positions when containers restart. AlgoPaca brings institutional quant rigor to personal accounts.

Traditional Retail SaaS Bots

THE STATUS QUO



Expensive SaaS Subscriptions (\$50 – \$300/mo)

Retail traders pay heavy recurring monthly fees before ever booking a single winning trade, eroding baseline alpha.



Black-Box "Trust Me" Execution

A bot that just says "BUY" demands blind faith. No indicator attribution, zero transparency, and no reasoning audit trail.



In-Memory State Fragility

Cloud bots hold stop orders in volatile memory. A crash or server restart abandons unhedged live positions without exit stops.



Siloed Technicals Without Macro Context

Simple indicator crossovers blindside traders during FOMC rate announcements, Treasury yield shifts, or earnings surprises.

△ Result: Disconnected accounts, unexpected liquidation, and subscription drag.

AlgoPaca Institutional Desk

THE QUANT SOLUTION



100% Free, MIT Open Source & \$0 Commissions

Zero software licensing fees, zero paywalls. Paired with Alpaca Markets for true commission-free US stock and ETF trading.



Full Multi-Modal AI Reasoning & Audit Trail

Synthesizes 5 technical indicators, live news sentiment, SEC earnings surprises, and macro yields into structured JSON with an explicit written thesis.



Crash-Safe Disk-Persisted Plans

Dip-hunt re-entries, buy-backs, and follow-on flip plans are committed to local SQLite disk per account. System restarts resume right where they left off.



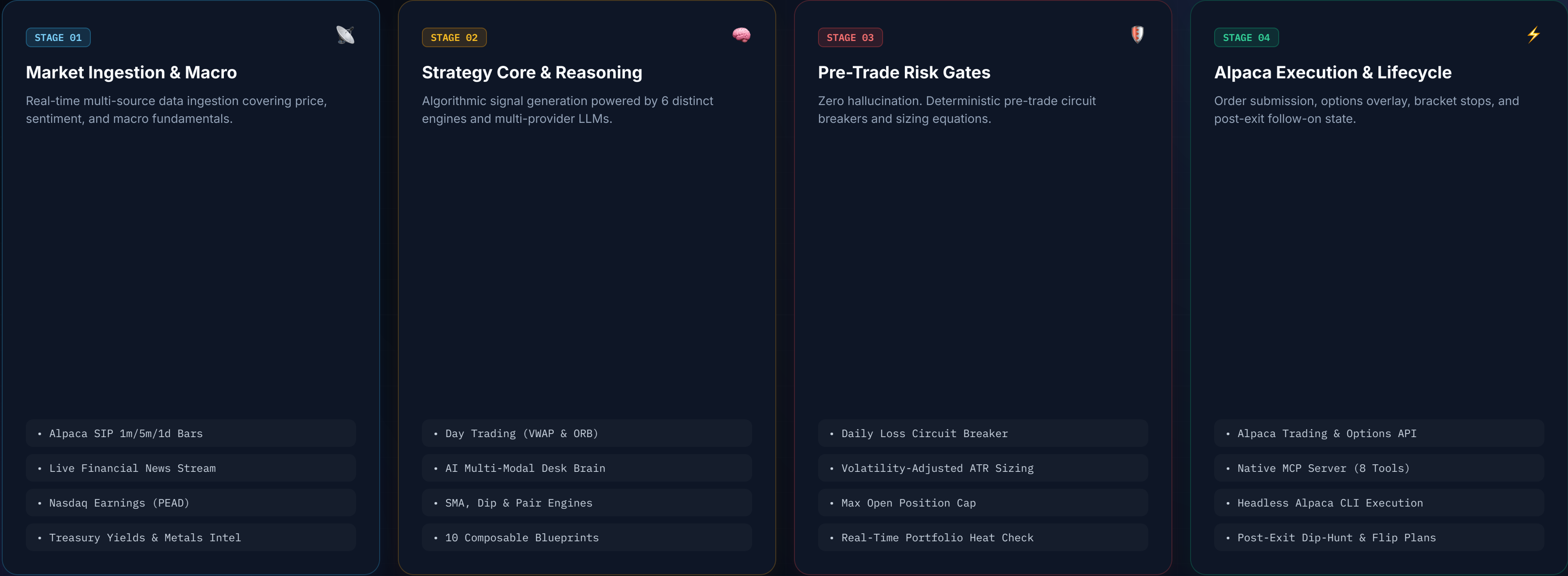
Macro Precious Metals & Cross-Asset Intelligence

Empirically derived cross-asset fusion blending Treasury yields, 1Y z-scored Gold-to-Silver Ratio (GSR), and the 200-day trend regime.

✓ Result: Sovereign control, institutional risk guardrails, and verified execution.

End-to-End Institutional Execution Pipeline

A deterministic four-stage lifecycle ensures every market tick passes through macro intelligence, quantitative strategy logic, strict mechanical risk filters, and fail-safe order execution.





6 Algorithmic Engines. 40+ Tuned Presets.

From sub-minute intraday scalping to macro swing rotation, AlgoPaca includes battle-tested quantitative engines ready to trade out-of-the-box or compose into custom blueprints.

1. Intraday Day-Trading

1M / 5M / 15M

Minute execution with VWAP Trend, 15m Opening Range Breakout (ORB) & Breakdown, and Momentum Scalps.

VWAP TREND 15M ORB EOD SQUARE-OFF

13 Presets • Long Only, Short Only, Two-Way (Long & Short)

2. AI Desk Brain

MULTI-MODAL QUANT

LLM synthesizes RSI, MACD, BB, ATR, ADX, live news, SEC earnings surprises, and macro metals into JSON actions.

BALANCED PEAD EARNINGS METALS MACRO

Confidence scoring • Human-approved prompt feedback loop

3. SMA Trend Following

TREND FOLLOWING

Moving average crossover filters across single names or watchlists with ATR trailing stops.

CLASSIC (10/30) FIBONACCI (8/21) GOLDEN CROSS

Includes Gold Momentum (50/150) & Custom moving average pairs

4. Quantitative Dip Hunter

MEAN REVERSION

Exploits capitulation washouts and recovery bounces using RSI washouts and lower Bollinger Band taps.

DEEP DIP MILD PULLBACK WASHOUT

Gold Bullion Dip Hunter preset for physical metal proxies

5. Pair & Regime Rotation

RELATIVE VALUE

Dynamic impulse rotation between 2 linked assets. Allocates to the winning regime or moves to cash.

GLD / SLV GLD / GDX CASH WHEN WEAK

Dynamic cash preservation when macro regimes break down

6. Regime Long/Short

DUAL MOMENTUM

Fast/slow EMA dual momentum with ADX trend strength gating. Holds through chop, exits on signal flip.

ADX FILTER DUAL EMA IS / OOS VALIDATION

Includes in-sample / out-of-sample backtest validation harness

Options Alpha Overlay Compatible: All 6 engines route signals into defined-risk vertical debit spreads, calls/puts, or covered hedges via Alpaca Options API.

10 Ready Blueprints • Custom Presets



Frontier & Open-Source AI. Structured Decision Output.

Switch seamlessly between Google Gemini, Anthropic Claude, OpenAI, and xAI Grok — with out-of-the-box compatibility for Featherless AI serverless open-source models (Llama-3, DeepSeek, Qwen). The AI synthesizes market telemetry into strict JSON schemas with confidence scores and written theses.

Google Gemini

SUB-SECOND

Models: 3.6 Flash, 3.5 Flash, 3.1 Pro
Extensive context windows for streaming financial news & earnings transcripts.

Anthropic Claude

DEDUCTIVE LOGIC

Models: 3.7 Sonnet, Opus 5, Haiku 4.5
Nuanced technical analysis & strict compliance with user risk guardrails.

OpenAI

DETERMINISTIC

Models: GPT-5.6 Sol / Terra, GPT-5.4
Industry-standard structured JSON output matching exact mathematical criteria.

xAI Grok

REAL-TIME MACRO

Models: Grok 4, Grok 4 Fast, Grok 3
High-speed social sentiment extraction & cross-market momentum detection.

Standardized Decision Contract

STRUCTURED JSON

```
{
  "symbol": "NVDA",
  "action": "BUY",
  "confidence": 0.82,
  "strategy": "ai_vwap_momentum",
  "entry_price": 124.50,
  "stop_loss_atr": 1.5,
  "target_rr": 2.0,
  "macro_bias": { "metals_score": +0.74, "yields_trend": "softening" },
  "thesis": "Clean 15m VWAP reclaim following positive Q3 datacenter revenue surprise. RSI (56) shows healthy accumulation with Treasury yields retreating from 4.30%."
}
```

ARCHITECTURAL LAW

The AI Veto Principle

The LLM acts as an **intelligent gatekeeper**, not an unchecked trader. It evaluates technical indicators and live context to decide whether to confirm or veto a trade.

✓ **Confidence Threshold Gate:** Set minimum confidence (e.g. 0.70). Lower scores are rejected before execution.

✓ **Can Veto, Never Invent:** The model can abort a questionable setup, but cannot fabricate random trades.

✓ **Zero Risk Rule Bypass:** An AI signal must still clear all 7 mechanical risk gates.

Deterministic calculations in Python • LLM handles semantic synthesis

Empirical Precious Metals & Macro Fusion

Real-time cross-asset data fusion across the full Alpaca metals universe. Factor weights are fitted strictly on measured 2007–2025 predictive power, not arbitrary market folklore.

Empirical Macro Weights

FITTED ON 2007–2025 DATA

1. Treasury Yield Direction (10Y / 2Y)	Weight: 0.40
2. 1-Year Z-Scored Gold-to-Silver Ratio (GSR)	Weight: 0.35
3. 200-Day Trend Regime (Moving Average Slope)	Weight: 0.15
4. US Dollar Index Momentum (UUP / DXY)	Weight: 0.10
5. Miner Relative Strength (GDX/GLD)	Weight: 0.00 (IC: -0.01)

💡 **Zero Folklore:** Miner strength is often touted in retail trading, but backtests revealed an Information Coefficient (IC) of -0.01. AlgoPaca assigns it exactly zero weight.

Statistical Return Monotonicity

SPEARMAN IC: +0.11

The resulting macro score demonstrates strict monotonic predictability on GLD forward 20-day returns across historical testing bands:

Most Bearish Score Band
High real yields, spiking DXY, elevated GSR
-0.29%

Neutral Regime Band
Balanced macro signals, normal volatility
+0.38%

Most Bullish Score Band
Falling yields, softening dollar, positive 200d MA
+1.14%

Universe: GLD, IAU, UGL, GLL, PHYS, SLV, AGQ, ZSL, SIL, SILJ, PSLV, GDX, GDXJ



Pre-Execution Risk Gates. Capital Preservation First.

Signals must survive 7 strict mechanical filters before an order reaches Alpaca. Configured out of the box to safeguard the \$100,000 hackathon paper account starting balance, with ATR volatility sizing and hard drawdown halts.

The 7 Pre-Trade Checkpoints

GATE 1	Daily Loss Circuit Breaker	Halt desk if Daily P&L $\leq$ -X%
GATE 2	Max Concurrent Open Positions	Hard cap on open inventory
GATE 3	Spread & Liquidity Sanity Gate	Rejects wide bid/ask anomalies
GATE 4	Post-Loss Cooldown Lockout	Anti-revenge cooldown after stop-out
GATE 5	Volatility-Adjusted ATR Sizing	$(\text{Equity} \times \text{Risk\%}) / (\text{ATR} \times \text{Mult})$
GATE 6	Max Portfolio Allocation Cap	Max single symbol allocation (e.g. 25%)
GATE 7	Real-Time Portfolio Heat Check	Total aggregate open risk across book

CORE DESIGN LAW

Bots are Refused. Humans are Advised.

An autonomous bot that trips any risk gate is refused outright — it logs the exact breach and halts. A human typing a ticket on the Advanced Order desk sees **every breach at once** with live portfolio heat, and must explicitly check an override before submission.

"The person is looking at the screen and may see what the rules cannot."

DYNAMIC SIZING EQUATION

Volatility-Adjusted Position Sizing

Shares = $(\text{Total Equity} \times \text{Risk\%}) / (\text{ATR}_{14} \times \text{Multiplier})$

Higher volatility automatically scales down share sizing. Quiet markets permit larger size while keeping the actual dollar risk completely constant across every setup.



Active Exit Desk & Crash-Safe Post-Exit Automation

Alpaca has no native "close-then-open" order class. AlgoPaca manages the complete order lifecycle locally, persisting armed post-exit plans to disk so restarts resume seamlessly.

1-Click Position Desk Exits

IN-TRADE PROTECTION



Quick Stop Loss Risk Chips

Attach stops instantly using predefined equity chips (1%, 2%, 3%, 5%, 8%, 10%) or custom dollar amounts.



Zero-Risk Breakeven Ratchet

One-click ratchet moves your stop to entry price ($\$entry \pm \0.01) immediately upon hitting +1.0R profit, securing a free roll.



Dynamic Trailing Stop Loss (%)

Follows the peak price upwards dynamically, locking in accumulated paper profits during explosive runs.



Partial Scaling with Live P&L Estimates

Scale out 25%, 50%, or 75% of your lot. The desk calculates estimated realized P&L before you confirm.

Includes Pre-Market & After-Hours Quote Mark-to-Market

Crash-Safe Post-Exit Plans

SURVIVES REBOOTS

1. Dip Hunt Re-Entry Plan

DIP BUYING

"If my stop fills, re-enter lower." Watches for a further dip % drop. Buys immediately if reached within timer; otherwise leaves a resting limit order.

2. Capital Re-Investment Plan

COMPOUNDING

"When this sell fills, put the cash back to work." Starts clock upon sell fill, recalculates available buying power, and buys back at target.

3. Next-Ticket Follow-On Flip

REGIME FLIP

"When this close fills, open opposite direction or rotate into symbol Y." Flip long→short or swap from gold into tech stocks automatically.

✓ Plans persisted to SQLite disk per account; a paper plan never executes live.



Desktop Power. Mobile-First PWA Freedom.

A lightning-fast Vanilla JS/CSS terminal with zero bloated dependencies. Complete with an interactive Auto-Trade approval queue, real-time push notifications, and multi-user isolation.



PWA READY

Mobile-First PWA Shell

Install directly to home screen on iOS & Android. Seamless touch interaction designed for on-the-go desk monitoring.

BOTTOM NAVIGATION BAR

[Desk] [Orders] [Auto-Trade] [Positions] [Menu]

- Sticky masthead ticker search
- 4 Themes (Obsidian, Slate, Emerald, Daylight)
- Multi-language: EN, BN, ES, FR, HI



HUMAN IN THE LOOP

Auto-Trade Approval Queue

Prefer oversight over 100% autonomous trading? Stage setups in the approval queue before submission.

NVDA • BUY • \$124.50

82% CONF

15m VWAP Trend Rider • ATR Stop: \$122.10

[Approve] [Reject]

- Web push browser notifications
- Branded HTML email alert dispatch
- 1-Click trade approval from phone



RBAC READY

Multi-User Isolation & RBAC

Enterprise isolation ready for prop desks or multiple traders. Strict role-based permissions and isolated workspaces.

ROLE HIERARCHY

Owner → Full system config, SMTP & users
Admin → Desk oversight & queue controls
Member → Personal account isolation

- User-scoped Alpaca & AI keys
- Active session & device revocation
- GDPR-compliant data export & purge



Deterministic Review. The Self-Improving Lessons Loop.

Every metric on the History Review is calculated mathematically in Python; the LLM is only asked to narrate the finished figures. Approved post-mortem lessons feed back into live prompts.

Deterministic Analytics First

ZERO HALLUCINATION



Engine P&L & Preset Attribution

Calculates realized win rates, profit factors, and dollar returns broken down by specific engine and preset.



Confidence Calibration Scoring

Statistically verifies if high-confidence entries (>0.80) actually produce higher win rates than lower confidence entries.



Hold Time & Execution Blocks

Audits average trade duration, distribution of stop-outs vs target limits, and reasons behind risk-blocked setups.



No Hallucinated Math: A hallucinated P&L on a quant trading desk is unacceptable. Python computes the numbers; the LLM provides prose narrative only.

The Human-Gated Lessons Loop

PROMPT FEEDBACK

STEP 1

Post-Mortem Trade Analysis

Compares entry thesis against real market outcome to pinpoint execution flaws.

STEP 2

Explicit Human Click Approval

A lesson only reaches the live bot if a human trader reads and explicitly approves it.

STEP 3

Injected into Live AI Prompts

Saved lessons are prepended into live strategy cycles for matching symbols.

// LIVE AI PROMPT INJECTION SAMPLE:

[APPROVED DESK LESSON for NVDA]: "Do not enter long breakout setups when overall tech sector breadth is negative and SPY is below its 15m VWAP."

Plus Walk-Forward Daily & Intraday Backtesting with Side-by-Side Run Comparison.



Alpaca Options Overlay & Dual Agent Interfaces (MCP + CLI)

Translate directional equity signals into defined-risk options contracts via Alpaca Options API, with dual agent execution: interactive tool-calling via Native MCP Server and headless automated execution via Alpaca CLI.

Alpaca Options Overlay Engine

DEFINED-RISK DERIVATIVES

Converts underlying equity setups into defined-risk options contracts with automated DTE target (21-45 days), delta targeting, and real-time Greek telemetry.

Vertical Debit Spreads

Bull Call & Bear Put spreads capping maximum capital loss strictly to the net debit premium paid.

Long Directional Options (ATM/OTM)

High-delta calls and puts for asymmetric directional breakout setups and earnings moves.

Covered Hedges & Income Writing

Protective puts and covered call overwrites against held equity positions to harvest Theta decay.

✓ Directly integrated into Alpaca Markets Options Chains & Trading API

Dual Agent Interfaces: MCP + CLI

MCP & CLI MODALITIES

Fulfills both hackathon modalities: Connect interactive AI assistants via Model Context Protocol, or drive headless execution pipelines with Alpaca CLI.

🔧 Native MCP Server (8 Tools)

CURSOR / CLAUDE

get_account • get_positions • run_strategy_cycle • place_order • run_backtest

📄 Headless Alpaca CLI Automation

CRON / CI / TERMINAL

alpaca order buy • alpaca position list • structured JSON output for long-running bots

"Claude, check our \$100k Alpaca paper portfolio, screen NVDA options chain, and deploy a Bull Call spread."

Run: ./run_mcp.sh • Seamless Claude Desktop, Cursor & Alpaca CLI Integration



One-Page Technical Summary: AI Logic, Risk Gates & Alpaca Stack

Official Hackathon Architecture Write-Up fulfilling all judging requirements for the Lablab.ai Alpaca AI Trading Agents Hackathon.

PILLAR 01 • AI CORE



AI Logic & Reasoning

- Multi-Provider Brain & Featherless**
Interchangeable engine supporting Gemini, Claude, GPT, Grok, and Featherless AI open-source financial models.
- Strict Decision JSON Contract**
Models ingest quantitative indicators (RSI, VWAP, ATR, MACD, BB) and return strictly validated actions with confidence ≥ 0.70 .
- The AI Veto Principle**
LLM serves as an intelligent gatekeeper: it can veto questionable setups based on macro/news, but can never hallucinate math or bypass risk rules.
- Closed-Loop Lessons Feedback**
Automated trade post-mortems generate lessons that, upon human click approval, are injected into future live agent prompts.

PILLAR 02 • RISK GATES



Deterministic Risk Gates

- 7 Pre-Execution Circuit Breakers**
Daily Loss Breaker, Max Open Positions, Spread Sanity, Post-Loss Cooldown, ATR Sizing, Allocation Cap, and Portfolio Heat Check.
- \$100,000 Portfolio Calibration**
Tuned for competition paper portfolios to strictly govern max single-ticket risk (1-2% equity) and eliminate margin calls.
- Volatility-Adjusted ATR Sizing**
Dynamically scales share/contract quantities inversely to market volatility, guaranteeing constant dollar risk across setups.
- Crash-Safe Persistence (SQLite)**
Post-exit automation plans (Dip Hunt Re-entry, Compounding Reinvestment, Regime Flip) survive server restarts without loss of state.

PILLAR 03 • ALPACA STACK



Alpaca Infrastructure

- Native Trading API & Market Data**
Full SIP multi-timeframe market bars, real-time WebSocket quotes, bracket orders, and account telemetry.
- Alpaca Options Derivatives**
Automated Options Chain screening with DTE targeting (21–45d), vertical debit spreads, long calls/puts, and covered hedges.
- Dual Agent Interface (MCP + CLI)**
Native Model Context Protocol server (8 tools) for Claude/Cursor plus Alpaca CLI automation for headless cron agent runs.
- Isolated Paper Trading Desk**
Zero real-money risk; verified execution against live market books in Alpaca's paper trading environment.

PUBLIC GITHUB REPO

github.com/mdjwel/algopaca

MIT License • 254 Tests Passing

LIVE DEMO PLATFORM

algopaca.spiderdevs.xyz

Full Web & Mobile PWA

HACKATHON ACCOUNT ID

Dedicated Paper Account

\$100,000 Starting Balance

60-SECOND DOCKER LAUNCH

`docker compose up -d`

Zero SaaS Fees • Self-Hosted